



Accelerated high-dimensional global optimization: A particle swarm optimizer incorporating homogeneous learning and autophagy mechanisms

Wen-Yuan Fu

College of Information Science and Engineering, Huaqiao University, 361002 Xiamen, China

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ABSTRACT

The curse of dimensionality often results in either premature convergence or slow convergence in high-dimensional global optimization. This paper proposes the high-speed homogeneous learning-based particle swarm optimizer (HLPPO) to address these challenges. The algorithm is based on the observation that individuals with similar attitudes or beliefs tend to coexist harmoniously in human societies, unlike birds or fish that squeeze or clash in the same space. By incorporating the concept of homogeneity into the optimization process, the algorithm controls particle learning and generates distinct subpopulations for competition and learning. Furthermore, to accelerate the convergence rate, the algorithm employs an autophagy mechanism and dynamic subpopulation diversity, guiding the population towards global optimal solutions while reducing redundant fitness evaluations. The effectiveness of HLPPO is illustrated through its exceptional performance in benchmark test suites, including the IEEE congress on evolutionary computation (CEC) 2008, CEC 2010, and CEC 2013, in both 1000 and 2000 dimensions. These results not only highlight its superiority over several state-of-the-art algorithms but also underscore its aptitude for effectively addressing low-dimensional optimization challenges encountered in CEC 2017.

1. Introduction

There are numerous complex optimization problems in engineering and science, and achieving rapid convergence to an optimal solution can significantly reduce computational costs. Conventional optimization techniques, including gradient-based methods, may not be appropriate for some real-world applications that lack gradient information or entail computationally demanding gradient calculations. In such cases, bio-inspired evolutionary algorithms that mimic biological evolution have proven to be effective optimization strategies because they do not require gradient information and possess a “black box” nature [1,2]. These algorithms have been successfully applied in a wide range of engineering applications, including convolutional neural network classifier [3], data clustering [4], video processing [5], and structural optimization [6]. Particle swarm optimization (PSO), a prominent flock dynamics technique developed by Kennedy and Eberhart [7], has received substantial interest in both industry and academia. PSO is based on the flock dynamics idea and is relatively straightforward to construct and implement. However, PSO is known to have sluggish convergence and have a tendency to converge to local optima [8]. Inappropriate parameter settings may potentially hinder the algorithm from reaching global convergence. As a result, enhancing PSO's global convergence is of great relevance.

E-mail address: fwy@hqu.edu.cn.

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Based on this, many PSO variants have been proposed, including hybrid PSO with fitness-distance balance and individual self-exploitation strategies (HPSO-FDB-ISE) [9], three-learning strategy PSO (TLS-PSO) [10], PSO with crisscross learning strategy (PSO-CL) [11], and adjustable driving force based PSO (ADFPSO) [12]. These PSO variants have improved global convergence and are capable of addressing both unimodal and multimodal optimization issues. However, when confronted with high-dimensional optimization problems, these PSO variations become inefficient [13] owing to the well-known “curse of dimensionality” phenomenon. It results in stronger interactions among variables across different dimensions, making it easy to get trapped in local optima [14,15]. Furthermore, the higher computational cost associated with large dimensionality has an effect on the convergence speed. This issue is particularly pronounced in recent years as high-dimensional optimization problems have become prevalent in various engineering applications, such as routing in very large scale integration circuit design [16], neural network training in large deep learning models [17], and cloud workflow scheduling [18]. As a result, for solving high-dimensional global optimization issues, an effective PSO algorithm that resists premature convergence and achieves rapid convergence is extremely desirable.

The large number of decision variables in the high-dimensional optimization problem can be defined as follows:

$$\min f(X), X = [x^1, x^2, \dots, x^D],$$

where $f(X)$ represents a continuous, nonlinear, and highly complex objective function, and X is a D -dimensional vector representing the decision variables of the optimization problem. As previously mentioned, the complexity of optimizing this problem using traditional PSO increases with D due to the difficulties involved in high-dimensional optimization [13–15].

A more straightforward approach for resolving high-dimensional optimization challenges is to break down the problem into smaller sub-functional optimization problems using the divide-and-conquer technique, which was first introduced by Potter and Jong as “cooperative coevolution” [19]. Despite its effectiveness in a variety of applications, cooperative coevolution still lacks robustness in the dimensional decomposition. Finding the best answer from the many alternative decompositions can be computationally difficult and time-consuming. Furthermore, it only works effectively for separable variable optimization issues and may perform poorly for strongly correlated or non-separable variable optimization problems. Multiple population coevolution is an alternate strategy that has been found to enhance optimization efficiency. This evolutionary method employs a fixed or variable number of populations, each with a size that may vary randomly or remain constant, to coevolve the optimal solution. However, because population size is not properly managed, this strategy may make it difficult to fully investigate valuable information during the evolution process. Hybrid evolutionary methods can also be used to improve convergence and provide a diverse set of information about the search space, such as combining reinforcement learning with PSO (RLPSO) [20], or using reinforcement learning in memetic PSO (RLMPSO) [21], or combining genetic algorithms with PSO (GL-PSO) [22]. However, because multiple optimization techniques are used, these methods can become more complex. To summarize, tackling high-dimensional global optimization problems remains a difficult task.

To address these potential challenges, this study proposes a high-speed homogeneous learning-based particle swarm optimizer (HLPSO) for high-dimensional optimization problems. The main contributions of this study are outlined as follows:

(1) HLPSO introduces a novel solution for high-dimensional optimization by incorporating the concept of homogenization learning to balance exploration and exploitation. This concept is realized through dynamically evolving multiple subpopulations based on homogenized competitiveness. Unlike traditional PSO, HLPSO enables the exchange of evolutionary information among subpopulations, enhancing search efficiency.

(2) A novel framework is introduced to assess population diversity, allowing for real-time adjustments to the evolutionary strategy. Importantly, the diversity measure focuses on subpopulations, dynamically varying its threshold.

(3) To expedite convergence in high-dimensional problems, a novel accelerated evolutionary strategy incorporating an autophagy mechanism is proposed. High-dimensional problems often involve numerous fitness evaluations, leading to redundant assessments and slower convergence. The proposed evolutionary strategy incorporates a cellular autophagy mechanism, utilizing the principle of self-removal of damaged cellular structures to eliminate redundant fitness evaluations and expedite convergence.

The rest of the paper is organized as follows: Section 2 dives into the technical foundations of combining PSO with high-dimensional optimization. In Section 3, the HLPSO algorithm is detailed, including its implementation and analysis. Section 4 presents the experimental test results, and Section 5 provides a summary and outlook to conclude the paper.

2. Related work

In this section, the fundamental concepts, general framework, and parameter settings of the canonical PSO involved in HLPSO are first introduced. Then, some related works on high-dimensional optimization based on PSO variants related to HLPSO are presented.

2.1. Canonical PSO

In the canonical PSO, each particle is regarded as a member of a group of solutions. The problem’s dimensionality is represented by D , and the search space is made up of N_p particles. The position of each particle, represented by P_i , is defined as a D -dimensional vector $X_i = [x_i^1, x_i^2, \dots, x_i^D]$, and its velocity is represented by a D -dimensional vector $V_i = [v_i^1, v_i^2, \dots, v_i^D]$, where $i = 1, 2, \dots, N_p$.

The PSO algorithm uses the following equations to update the position and velocity of each particle:

$$V_i = \omega \cdot V_i + c_1 r_{1i} (Pb_i - X_i) + c_2 r_{2i} (Pg - X_i) \quad (1)$$

$$X_i = X_i + V_i \quad (2)$$

where ω represents the inertia weight. The position and velocity of particle i in the population are represented by X_i and V_i , respectively. The acceleration coefficients are represented by c_1 and c_2 . r_{1i} and r_{2i} are random variables drawn from the interval (0,1). The best historical position of particle i is represented by $Pb_i = [pb_i^1, pb_i^2, \dots, pb_i^D]$, and the population's global best position is represented by $Pg = [pg^1, pg^2, \dots, pg^D]$. If the acceleration coefficient is set too low, particles may hover around the search area. If it is set too large, the particles may rapidly converge on or even overshoot the search area. Furthermore, because the value V_i is too large or too low, the calculation process of equation (1) may cause the position parameter X_i to exceed the solution space. To ensure the validity of the search solution, the range of velocity and position must be limited. Each particle's velocity and position in each dimension must be within the bounds $[V_{\min}^j, V_{\max}^j]$ and $[X_{\min}^j, X_{\max}^j]$, respectively, where $j = 1, 2, \dots, D$.

2.2. PSO variants for high-dimensional optimization

Recent research indicates that PSO variants have been used to solve small-scale optimization problems [9,10,12]. However, as the problem dimensionality increases, the number of local extrema points also increases, leading to a greater computational burden and a less efficient optimization process [8,23].

In the realm of high-dimensional optimization research, two primary methods have been adopted to address these challenges: cooperative coevolutionary frameworks and hybrid evolutionary frameworks. One of the prominent algorithms in the cooperative coevolutionary framework is the cooperative coevolution PSO (CCPSO). CCPSO is based on the PSO architecture and has several variations, such as cooperative coevolution PSO with SK model (CCPSO-SK) and CCPSO with HK model (CCPSO-HK) [24,25]. CCPSO-HK [24] combines CCPSO-SK and canonical PSO to update the population. Another notable algorithm in this framework is the differential evolution-based cooperative coevolution group (DECC-G) [26], which has demonstrated effectiveness in addressing high-dimensional optimization problems. Additionally, various techniques, such as the differential grouping technique (DG) [15], DG2 [27], multilevel cooperative coevolution (MLCC) [28], eXtended DG [29], and covariance matrix adaptation evolution strategy (CMAES) [30], have been proposed to identify interdependent variables. Despite these efforts, these techniques may still struggle with optimal dimensional decomposition and convergence performance in cases involving closely coupled interactions or non-separable variables. The computation time for dimension decomposition also increases with problem dimensionality, leading to slower convergence rate.

To overcome the limitations of cooperative coevolutionary frameworks, novel hybrid search techniques based on PSO have been developed to enhance global convergence performance in high-dimensional optimization. For instance, reference [18] proposed a dynamic multi-swarm PSO (DMS-L-PSO) that randomly varies the population size per iteration. However, the random setup of DMS-L-PSO can make it difficult to achieve the desired population size, and the stochastic alteration mechanism may not effectively gather information. On the other hand, reference [31] introduced a fully informed PSO (FIPS), which updates each particle based on the historical best positions of multiple neighbors. Reference [32] presented a social learning PSO (SL-PSO), where particles learn from each other based on their fitness rankings, and this concept was advanced in [33] with the development of competitive swarm optimization (CSO), in which two particles compete and the loser learns from the winner. SL-PSO and CSO use superior particles and the global mean position of all particles to steer evolution, rather than just pbest and gbest. In [34], a sinusoidal function was used to balance convergence performance by adjusting particle learning probability. Reference [35] improved particle escape from local optima by incorporating simulated annealing into PSO, and presented a novel method for designing a more efficient covering array. Reference [36] introduced a novel two-stage metaheuristic method that combines comprehensive learning PSO (CLPSO) [37] and evolutionary structural optimization, resulting in improved convergence performance for designing standard section steel structures. Reference [38] developed dynamic learning-based swarm optimization (DLLSO), which divides particles into levels and guides each particle to a higher level using two particles from different levels. However, DLLSO may limit update procedures for promising particles and hinder diversity maintenance. Other hybrid evolutionary methods such as segment-based predominant learning swarm optimization (SPLSO) [39], two-phase learning-based swarm optimization (TPLSO) [40], RLLPSO [20], heterogeneous cognitive learning PSO (HCLPSO) [41], three-phase co-evolutionary approach (TPCSO) [42], and GL-PSO [22] have also been proposed to address premature convergence. Despite these efforts, premature convergence and slow convergence rates remain major challenges in high-dimensional optimization.

3. HLP SO

This section introduces HLP SO, a variant of PSO designed to improve the global performance of high-dimensional optimization. The proposed method has four components. The first section introduces the concept of homogeneity in the evolutionary process, which aims to strike a balance between exploration and exploitation. In the second section, a novel framework for evaluating population diversity is presented, which is essential for high-dimensional optimization. The third section describes a novel evolutionary strategy that accelerates convergence rate via an autophagy mechanism. Finally, the computational complexity analysis of HLP SO is presented, shedding light on its efficiency.

3.1. The homogeneous learning concept

In contrast to birds and fish, human movement is not always unified or coordinated. Instead, people in human societies tend to form homogeneous social groups with those who share similar beliefs and attitudes. This is evidenced by the fact that in human

societies, two individuals with similar beliefs and attitudes rarely clash, whereas two birds or two fish in the same space may squeeze or clash. In the field of high-dimensional optimization, the concept of homogeneous learning can be incorporated into various evolutionary processes in order to regulate the learning capabilities of subpopulations and maintain a balance between exploration and exploitation. This serves as the foundation for the HLPPO algorithm.

Assume the population size is N_p and the optimization problem has D dimensions. Using a homogenized competition strategy, the particles in the population are divided into three subpopulations in HLPPO. The first subpopulation, M_1 , contains positive particles (X_P) with the best fitness values, the second, M_2 , contains homogenized particles (X_H) with the second-best fitness values, and the third, M_3 , contains negative particles (X_N) with the worst fitness values. These subpopulations adjust their positions in response to competition and learning. Equations (3) and (4) give the update law used to calculate the position and velocity of negative particles X_N in subpopulation M_3 .

$$V_i = r_{1i} V_i + r_{2i} (X_{Pi} - X_{Ni}) + r_{3i} (\bar{X}_P - X_{Ni}) + \eta \cdot r_{4i} (\bar{X}_H - X_{Ni}) \quad (3)$$

$$X_{Ni} = X_{Ni} + V_i \quad (4)$$

Here, r_{1i} , r_{2i} , r_{3i} , and r_{4i} are uniformly distributed random variables drawn from the interval (0,1). The variables X_{Pi} and X_{Ni} correspond to the i -th positive particle in subpopulation M_1 and the i -th negative particle in subpopulation M_3 , respectively. The mean positions of positive particles in subpopulation M_1 and homogenized particles in subpopulation M_2 are represented by $\bar{X}_P$ and $\bar{X}_H$, respectively. η is a control parameter with a value in the range [0,1]. As inspired by CSO, positive particles are considered dominant, actively leading the population forward. Negative particles, on the other hand, are thought to be weak and must learn useful information from other particles in order to survive. To maintain diversity, the negative particles learn from the positive particles and gather information from the mean position of the dominant population. In the evolution process of particle competition and classification, homogenized particles are neither excellent positives nor poor negatives. Excellent particle diversity is generally low, whereas poor particle diversity is relatively high, resulting in less exploration for positive particles and less exploitation for negative particles. Homogenized particles are intermediate particles with different properties that contain more information about potential convergence and diversity. By incorporating their mean position as an update term, particles can gain more useful insights to guide the algorithm's evolution.

3.2. Diversity identification

This paper introduces a novel framework for identifying diversity that seeks both exploratory and exploitative behaviors. To effectively quantify diversity in an evolutionary process, this paper presents a novel framework that embraces both exploratory and exploitative behaviors. The diversity measure, D_M , is articulated through the equation below:

$$D_M = \sum_{i=1}^{\lceil N_p/3 \rceil} \sum_{j=1}^D |x_i^j - \bar{x}_i^j| \quad (5)$$

where $\bar{x}_i^j$ represents the mean position of the i -th particle in the j -th dimension and x_i^j represents the position of the i -th particle in the j -th dimension. The symbol $\lceil \cdot \rceil$ represents the ceiling function.

However, traditional diversity measures often suffer from a critical limitation—they heavily rely on predefined thresholds. When diversity exceeds these thresholds, certain operators are activated to decrease diversity; conversely, when diversity falls below a certain level, other operators are engaged to increase it, as demonstrated in [33]. Unfortunately, the need for threshold setting can introduce ambiguity and complexity, particularly when dealing with different optimization problems.

In contrast, this paper introduces an innovative approach to diversity measurement that shifts the focus to subpopulations, offering a dynamic method for identifying diversity during the evolutionary process. Specifically, this method is described by the following equations:

$$D_{M_1} = \sum_{i=1}^{\lceil N_p/3 \rceil} \sum_{j=1}^D |x_{Pi}^j - \bar{x}_{Pi}^j| \quad (6)$$

$$D_{M_2} = \sum_{i=1}^{\lceil N_p/3 \rceil} \sum_{j=1}^D |x_{Hi}^j - \bar{x}_{Hi}^j| \quad (7)$$

$$D_{M_3} = \sum_{i=1}^{\lceil N_p/3 \rceil} \sum_{j=1}^D |x_{Ni}^j - \bar{x}_{Ni}^j| \quad (8)$$

where D_{M_1} , D_{M_2} , and D_{M_3} represent the diversity of the positive particle population M_1 , homogenized particle population M_2 , and the negative particle population M_3 , respectively. Utilizing the diversity measures of subpopulations, the algorithm gains an insightful perspective on the state of population evolution. By quantifying the relative magnitudes of these measures, the algorithm can dynamically adapt its optimization strategy, as discussed further in the subsequent section.

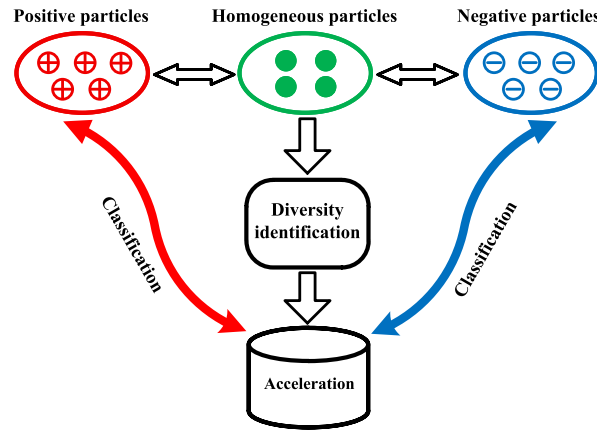


Fig. 1. The framework of HLPPO.

3.3. An accelerated evolutionary strategy with an autophagy mechanism

An evolutionary strategy that incorporates autophagy may speed up convergence. In the pursuit of enhancing convergence speed, an evolutionary strategy can harness the power of autophagy. High-dimensional optimization problems, often characterized by their large scale, frequently exhibit intricate interplay between dimensions, which can lead to redundant function evaluations and consequently impede convergence.

The autophagy mechanism is activated if the diversity of subpopulation M_3 lags behind that of M_1 . To achieve this, the activation of the autophagy mechanism is contingent upon the relative diversity levels of subpopulations, specifically M_3 and M_1 . When diversity of M_1 exceeds that of M_3 , the positive particle is prioritized. The autophagy mechanism removes the second and third terms of equation (3), yielding the following algorithm:

$$V_i = r_{1i} V_i + r_{2i} (X_{Pi} - X_{Ni}) \quad (9)$$

Conversely, if the diversity of subpopulation M_3 lags behind that of M_2 , it indicates the active participation of homogenizing particles in driving the evolution toward the global optimum. As a result, the homogenizing particle exerts a strong influence, and the autophagy mechanism eliminates the first and third terms of equation (3). Based on this observation, a novel, accelerated evolutionary strategy known as the autophagy mechanism (equations (9) and (10)) is developed:

$$V_i = r_{1i} V_i + \eta \cdot r_{3i} (\bar{X}_P - X_{Ni}) \quad (10)$$

The HLPPO framework, depicted in Fig. 1, combines the homogeneous learning strategy, diversity identification, and accelerated evolution strategy with autophagy described above. Additionally, Algorithm 1 summarizes the HLPPO pseudocode.

This HLPPO execution process unfolds through three key steps, as shown in Fig. 1. The population is first grouped, and three particles are chosen at random from the population M . Following that, the particles are sorted based on their fitness values, with the highest-performing particle designated as the positive particle and placed in M_1 , the homogenized particle designated as the second-best-performing particle and placed in M_2 , and the particle with the lowest performance designated as the negative particle and placed in M_3 . This classification procedure is repeated until all particles in the population M are assigned to M_1 , M_2 , or M_3 , which corresponds to Algorithm 1 lines 4 to 8. The diversity of the subpopulations is identified in the second step. Equations (6)–(8), which correspond to line 10 of Algorithm 1, can be used to calculate the diversity of each corresponding subpopulation D_{M_1} , D_{M_2} , and D_{M_3} . The third step is the acceleration process, in which the autophagy mechanism speeds up the algorithm's convergence. Lines 11 to 19 of Algorithm 1 correspond to this acceleration process, while line 20 corresponds to the position update rule of the subpopulation M_3 .

3.4. The computational complexity analysis of HLPPO

The time and space complexity of HLPPO are discussed in this section. Because HLPPO is essentially a variant of PSO, it retains the time complexity of PSO. It is important to note that the fitness evaluation time varies depending on the optimization problem, so the time complexity analysis of evolutionary algorithms does not include it. Lines 5 and 6 of Algorithm 1 state that it takes $O(N_p \log(N_p) + N_p)$ to competitively divide and rank the populations. Furthermore, as described in lines 10–19 of Algorithm 1, the particles in the population are updated in batches of $\lceil N_p/3 \rceil$ at a time. This operation has a time complexity of $O(N_p/3 \times D)$.

In terms of space complexity, HLPPO requires less space than conventional PSO, which occupies $O(N_p \times D)$ space. This is due to the fact that HLPPO does not store the individual best position of each particle, but rather computes it on-the-fly while the algorithm is running. Because of the reduction in space complexity, HLPPO is better suited for high-dimensional optimization problems with limited memory resources.

Algorithm 1 The pseudocode for the HLPPO algorithm. N_p denotes the size of population M . The maximum number of fitness evaluations constitutes the terminal condition.

```

1: Initialize population  $M$  randomly and calculate their fitness value;
2: while the terminal condition (e.g. maximum number of fitness evaluations) is not satisfied do
3:   //Classification
4:   for  $k = 1, 2, \dots, \lceil N_p/3 \rceil$  do
5:     Three particles are randomly chosen from the population  $M$ . They are then ordered based on their fitness
       value, with the highest-performing particle referred to as the positive particle and placed in  $M_1$ , the second-
       best referred to as the homogenized particle and placed in  $M_2$ , and the lowest-performing referred to as the
       negative particle and placed in  $M_3$ ;
6:      $|M| \leftarrow |M| - 3$ ;
7:   end for
8:    $M \leftarrow M_1 \cup M_2 \cup M_3$ ;
9:   //Diversity identification
10:  Calculate  $D_{M_1}$ ,  $D_{M_2}$  and  $D_{M_3}$  according to equation (6)–(8), respectively;
11:  if  $D_{M_3} < D_{M_1}$  then
12:    //Acceleration
13:    update  $V_i$  using equation (9);
14:    if  $D_{M_3} < D_{M_2}$  then
15:      update  $V_i$  using equation (10);
16:    else
17:      update  $V_i$  using equation (3);
18:    end if
19:  end if
20:  update  $X_{N_i}$  using equation (4);
21: end while

```

4. Experimental studies

The performance of HLPPO is validated in this section by using 70 benchmarks from the CEC 2008 [43], CEC 2010 [44], CEC 2013 [45], and CEC 2017 [46] test suites. These benchmarks are categorized into separable (F_{a1} , F_{a4} , F_{a6} , F_{b1} – F_{b3} , F_{c1} – F_{c3} , F_{d1} – F_{d3}), partially separable (F_{b4} – F_{b18} , F_{c4} – F_{c11} , F_{d4} – F_{d10}), inseparable (F_{a2} , F_{a3} , F_{a5} , F_{b19} , F_{b20} , F_{c15}), hybrid (F_{d11} – F_{d20}), and composite (F_{d21} – F_{d30}) functions. The CEC 2017 test suite employs 29 functions, with the exception of F_{d2} , which is excluded due to its high randomness in function values. It is noteworthy that the CEC 2013 test suite includes new features such as the imbalance contribution of sub-components and overlapping functions, which make them more complex and difficult to optimize than the CEC 2010 functions. The results of TLS-PSO [10], ADFPSO [12], PSO-CL [11], and HPSO-FDB-ISE [9] are taken from their respective papers, and the rest of the algorithms are implemented using MATLAB (2020b) on a PC equipped with a 2.20 GHz Intel(R) Xeon(R) Silver 4114 processor, 32 GB memory, and Windows 10 64-bit operating system. 30 independent runs are performed to ensure thorough testing and avoid overfitting. The CEC 2008, CEC 2010, and CEC 2013 test suites are used in this study for large-scale function optimization, with a maximum fitness evaluation of $3000 \times D$, whereas the CEC 2017 test suite is designed for low-dimensional optimization, with a maximum fitness evaluation of $10000 \times D$, where D is the dimension of the optimization problem. Nonparametric tests are also used to determine experiment statistical significance. This section investigates the HLPPO design parameters and autophagy mechanism, compares it to other excellent PSO and non-PSO variants, and investigates the convergence performance of 2000-dimensional and low-dimensional optimization problems.

4.1. Design parameter study

The HLPPO makes use of two design parameters: the particle population size N_p and the control parameter η . To ensure the effectiveness of HLPPO's design, a comprehensive study is conducted to optimize these crucial design parameters. Finding optimal values for these parameters, on the other hand, can be difficult. A small N_p can lead to premature convergence and accelerated information interaction among particles, whereas a large N_p can weaken information interaction and increase the computational cost of information transfer. In this study, experiments are carried out to optimize N_p and η , with N_p ranging from 450 to 720 and η ranging from 0.70 to 1.00.

Table 1 presents statistical results for various combinations of N_p and η in HLPPO for eight benchmarks from CEC 2010, including partially separable and multimodal functions (F_{b6} , F_{b11} , F_{b13} , and F_{b18}) as well as unimodal separable functions (F_{b1} and F_{b2}) and partially separable unimodal functions (F_{b9} and F_{b12}). Partially separable and multimodal functions are regarded as more difficult and representative of industrial optimization problems. The results show that unimodal functions performed better with N_p values of 450 and 510, while multimodal functions performed better with N_p values of 600 and 720. A smaller N_p does not provide sufficient diversity for the swarm, whereas a larger N_p can result in improved convergence performance for multimodal problems. Conversely, since η controls the degree of homogeneous learning in HLPPO, particles with homogeneous characteristics contain more information about potential convergence and diversity. Using their average position as an update term enables the particles to gain useful insights to guide the evolution. According to equation (3), the impact of positive particles becomes more pronounced as η decreases. Positive particles are generally the best-performing particles with the least diversity, and when combined, HLPPO tends to fall into local optima in this scenario. When η is set to 0.70, the optimization performance suffers. To compensate for the lack of diversity, η needs

Table 1

The mean and standard deviation (std) values (in parentheses) of the fitness values obtained by HLPPO in the benchmark of CEC 2010 1000- D , with population size N_p ranging from 450 to 720 and η from 0.70 to 1.00.

N_p	Fun	$\eta = 0.70$	$\eta = 0.90$	$\eta = 0.95$	$\eta = 1.00$
$N_p = 450$	F_{b1}	4.58E+04(8.85E+03)	6.82E-19(3.54E-20)	6.13E-19(4.05E-20)	5.45E-19(3.46E-20)
	F_{b2}	3.05E+03(3.06E+02)	2.95E+03(8.26E+01)	1.74E+03(8.67E+01)	1.11E+03(8.16E+01)
	F_{b6}	2.35E+01(1.05E-01)	3.24E-06(2.47E-07)	1.54E-06(3.14E-08)	1.27E-07(4.48E-09)
	F_{b9}	1.38E+06(2.67E+05)	7.85E+06(7.56E+02)	7.08E+06(7.28E+02)	6.63E+06(5.85E+02)
	F_{b11}	3.59E+01(6.59E-01)	2.81E+01(7.83E-01)	2.87E+01(7.81E-01)	1.93E+01(7.17E-01)
	F_{b12}	6.03E+04(4.36E+03)	3.86E+02(2.83E+02)	3.57E+02(2.54E+02)	1.59E+02(1.46E+02)
	F_{b13}	1.18E+03(6.63E+01)	9.82E+01(8.95E+00)	9.48E+01(9.24E+00)	9.95E+01(8.50E+00)
	F_{b18}	2.18E+03(4.31E+02)	6.19E+02(2.02E+02)	6.28E+02(1.89E+02)	5.91E+02(1.17E+02)
$N_p = 510$	F_{b1}	2.13E+04(6.12E+03)	8.15E-19(9.25E-20)	6.02E-19(8.16E-20)	6.47E-19(4.31E-20)
	F_{b2}	2.25E+03(1.43E+02)	1.75E+03(6.27E+01)	1.13E+03(4.85E+01)	1.03E+03(5.12E+01)
	F_{b6}	1.96E+01(2.25E-02)	4.11E-07(1.92E-09)	3.05E-07(2.58E-09)	1.23E-07(3.94E-09)
	F_{b9}	1.23E+06(6.23E+05)	8.14E+06(8.65E+02)	7.25E+06(8.23E+02)	6.13E+06(5.41E+02)
	F_{b11}	1.02E+01(3.62E-01)	1.95E+01(4.10E-01)	1.81E+01(4.05E-01)	1.01E+01(9.36E-01)
	F_{b12}	6.58E+04(5.11E+03)	1.23E+02(2.08E+02)	1.54E+02(2.15E+02)	1.32E+02(1.26E+02)
	F_{b13}	2.27E+03(9.13E+01)	1.01E+02(1.35E+01)	1.03E+02(1.26E+01)	9.91E+01(9.95E+00)
	F_{b18}	2.09E+03(3.57E+02)	7.85E+02(2.84E+02)	7.28E+02(1.35E+02)	6.92E+02(1.67E+02)
$N_p = 600$	F_{b1}	3.09E+04(7.94E+03)	8.23E-19(5.22E-20)	8.05E-19(1.34E-20)	7.91E-19(2.59E-20)
	F_{b2}	2.01E+03(9.70E+01)	1.22E+03(9.42E+01)	1.14E+03(2.95E+01)	9.15E+02(5.62E+01)
	F_{b6}	1.96E+01(2.25E-02)	1.95E+01(4.37E-02)	4.38E-05(6.10E-06)	1.33E-07(3.20E-09)
	F_{b9}	4.86E+07(3.47E+06)	5.64E+06(3.26E+05)	5.33E+06(3.42E+05)	6.20E+06(6.08E+05)
	F_{b11}	3.94E+01(7.87E-01)	1.98E-11(1.45E-12)	1.97E+01(1.30E-01)	2.10E-12(1.53E-12)
	F_{b12}	8.37E+04(5.27E+03)	1.81E+03(5.85E+02)	4.95E+02(3.88E+02)	8.15E+01(2.63E+01)
	F_{b13}	4.08E+03(8.29E+01)	2.92E+02(1.83E+01)	1.15E+02(2.26E+01)	1.80E+02(1.34E+02)
	F_{b18}	2.72E+03(4.20E+02)	1.72E+03(3.81E+02)	1.02E+03(3.38E+02)	6.49E+02(1.28E+02)
$N_p = 720$	F_{b1}	3.52E+04(8.62E+03)	5.78E-18(5.49E-19)	2.45E-18(2.54E-19)	1.15E-18(1.32E-20)
	F_{b2}	2.35E+03(9.82E+01)	9.58E+02(3.57E+01)	8.25E+02(2.85E+01)	8.34E+02(1.54E+01)
	F_{b6}	1.23E+01(1.58E-02)	3.82E-06(7.25E-7)	2.45E-06(3.78E-8)	1.75E-07(5.98E-10)
	F_{b9}	4.25E+07(3.21E+06)	4.53E+06(3.11E+05)	5.82E+06(2.98E+05)	5.60E+06(1.09E+05)
	F_{b11}	3.28E+01(6.28E-01)	1.12E+01(2.06E-01)	3.52E-12(2.57E-13)	2.32E-12(1.18E-13)
	F_{b12}	8.54E+04(5.14E+03)	2.31E+01(2.88E+01)	1.85E+01(2.62E+01)	1.50E+01(1.16E+01)
	F_{b13}	3.98E+03(8.10E+01)	2.56E+02(3.87E+01)	1.24E+02(3.54E+01)	1.19E+02(2.07E+01)
	F_{b18}	2.18E+03(2.18E+02)	1.05E+03(7.76E+01)	9.05E+02(8.71E+01)	7.54E+02(7.09E+01)

to be increased in the case of decreasing N_p . When using HLPPO for high-dimensional optimization, the optimal values for N_p and η are 510 and 1.0, respectively.

4.2. Effects of autophagy mechanism

The proposed autophagy mechanism in HLPPO is validated by comparison with TPLSO [40], CSO [33], and GPSO [7]. The test results in Fig. 2 demonstrate that the proposed mechanism has the potential to improve the HLPPO's convergence speed when compared to TPLSO and CSO using the CEC 2008 test suites.

Furthermore, when changes in population diversity are detected, the autophagy mechanism is activated, accelerating convergence speed. This is seen in the convergence process of HLPPO, TPLSO, CSO, and GPSO for F_{a1} - F_{a6} in the CEC 2008 benchmarks, as shown in Fig. 2. Notably, the proposed homogeneous learning strategy achieves superior convergence performance for separable functions F_{a1} and F_{a4} , as well as the unimodal function F_{a2} . Fig. 2's subplots (a), (b), and (d) demonstrate that F_{a1} , F_{a2} , and F_{a4} always have a faster convergence speed. Fig. 2 subplots (c), (e), and (f) show that, despite its slower initial convergence speed, HLPPO gradually outperforms the other three PSO variants during the middle and later stages of the convergence process. It should be noted that in the optimization of F_{a6} , both HLPPO and CSO exhibit extremely rapid convergence, leading to the coordinate axes being unable to display the final convergence results. The results in Fig. 2 and Table 2 indicate that the autophagy mechanism causes HLPPO to converge more precisely and quickly.

In conclusion, the proposed autophagy mechanism in HLPPO has been shown to improve the convergence rate. This is accomplished by activating the mechanism only when changes in population diversity are detected, allowing for more efficient exploration and exploitation of the search space. Fig. 2 shows that in the CEC 2008 benchmarks, HLPPO outperforms TPLSO, CSO, and GPSO in terms of convergence accuracy and rate, especially for separable and unimodal functions.

4.3. Comparison with state-of-the-art algorithms for PSO variants

To fully assess the performance of the proposed HLPPO, two groups of state-of-the-art high-dimensional optimization algorithms are compared, namely PSO and non-PSO variants. This subsection first compares HLPPO to other PSO variants, and the following

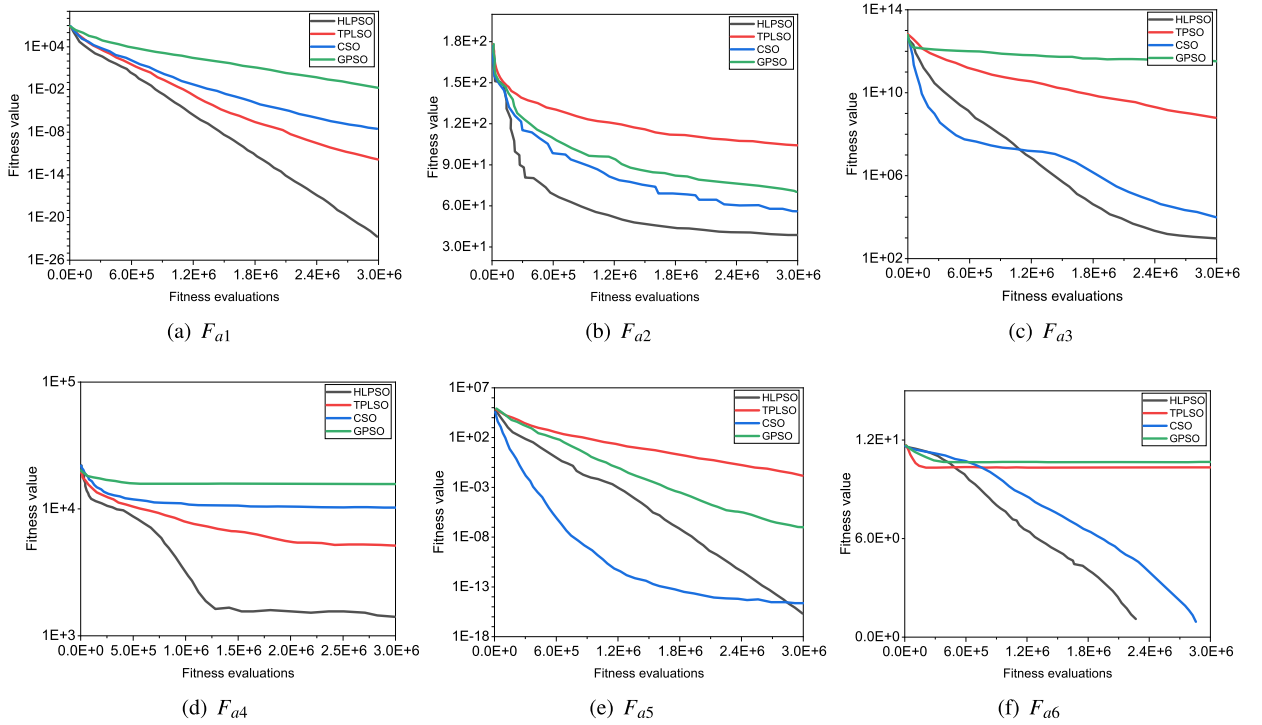


Fig. 2. Fitness value comparison among HLPso, TPLso, CSO, and GPSO on 1000-D CEC 2008 benchmarks F_{a1} - F_{a6} with $3.0E+6$ fitness evaluations.

subsection compares HLPso to other non-PSO variants. The CEC 2010 and CEC 2013 test suites are used, both with 1000 dimensions. As comparison methods, eight popular PSO variants are chosen: DLLSO, TPLso, AGLDPSO, CSO, SL-PSO, DMS-L-PSO [18], RLLPSO [20], and SinSLSO [34]. The key parameters of each algorithm are set as suggested in their respective papers to ensure a fair comparison. Tables 2 and 3 present the experimental results for the CEC 2010 and CEC 2013 test suites, including the mean, standard deviation, and two-tailed Wilcoxon rank sum test with the significance level $\alpha = 0.05$. The final row of the table compares HLPso's and the comparison algorithm's convergence performance, where "+" indicates that HLPso is significantly better than the comparison algorithm, "-" indicates that HLPso is significantly worse, and "=" indicates that there is no significant difference between HLPso and the comparison algorithm. Moreover, "+/-/" in the table's last row indicates the number of wins, losses, and draws obtained by HLPso and the comparison algorithm on different test functions.

Table 2 illustrates that HLPso outperforms the comparison algorithm for the majority of the CEC 2010 test suite's 20 benchmarks. When compared to SL-PSO, HLPso only loses on F_{b10} and wins on the rest 19 benchmarks. When compared to TPLso and RLLPSO, HLPso loses on two benchmarks, F_{b1} and F_{b2} , but wins on the remaining 11 benchmarks. Notably, F_{b1} and F_{b2} are separable functions that require less diversity information for optimization and so are easier to optimize than partially separable and entirely non-separable functions. HLPso also beats six popular PSO variants on six benchmarks: F_{b4} , F_{b9} , F_{b13} , F_{b18} , F_{b19} , and F_{b20} , while failing on five: F_{b7} , F_{b10} , F_{b11} , F_{b15} , and F_{b16} . In comparison to the best performing of the eight PSO variants, TPLso, DLLSO, AGLDPSO, RLLPSO, and SinSLSO, HLPso only fails on one partially separable unimodal function (F_{b7}) and four partially separable multimodal functions (F_{b10} , F_{b11} , F_{b15} , and F_{b16}). Furthermore, HLPso's convergence performance is comparable to that of DLLSO on F_{b2} and AGDPSO on F_{b8} . Overall, HLPso beats the popular PSO variants on most partially separable and non-separable functions, especially on the fully non-separable functions F_{b19} and F_{b20} , which are particularly difficult to handle. This shows that HLPso has greater diversity than DLLSO, AGDPSO, TPLso, RLLPSO, and SinSLSO.

The CEC 2013 test suite is more difficult to optimize due to the fact that it has the asymmetric and non-regular nature of real-world industrial optimization problems. Regarding the results of CEC 2013, the statistics in Table 3 show that HLPso wins on nine out of 15 benchmarks and comes in second place on the F_{c2} and F_{c9} functions. Specifically, HLPso wins on two of the three separable benchmarks F_{c1} - F_{c3} , five of the eight partially separable benchmarks F_{c4} - F_{c11} , and two of the eight non-separable functions F_{c12} - F_{c15} . The best averages obtained by the other PSO variants are all less than or equal to two functions, with RLLPSO obtaining the best averages on F_{c7} and F_{c15} . In particular, SinSLSO outperforms HLPso on eight benchmarks in CEC 2010, but in the CEC 2013 test suite, HLPso outperforms SinSLSO on all but one benchmark (F_{c11}). According to these findings, HLPso is a formidable optimization algorithm that can be applied to a wide variety of practical optimization problems, especially those characterized by asymmetry and irregularity.

Table 2

Experimental statistics of HLPPO and other PSO variants in the 1000-dimensional CEC 2010 benchmark test suite. The best average results for each benchmark are highlighted in bold.

Fun	HLPPO	DLPO	AGLDPPO	CSO	TPLPO	SL-PSO	DMS-L-PSO	RLLPO	SinSLPO
	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)
F_{b1}	6.47E-19 (4.31E-20)	2.97E-22 (1.27E-22)-	2.06E-21 (3.15E-22)-	4.56E-12 (7.16E-13)+	4.13E-19 (2.18E-20)=	7.35E-18 (2.37E-18)+	1.58E+08 (5.04E+06)+	4.02E-22 (2.03E-22)-	2.34E-21 (2.65E-22)-
F_{b2}	1.03E+03 (5.12E+01)	1.01E+03 (5.27E+01)=	2.24E+03 (8.25E+02)+	7.85E+03 (3.42E+02)+	8.32E+02 (5.21E+01)-	1.96E+03 (2.17E+02)+	6.52E+03 (5.28E+02)+	9.35E+02 (7.08E+01)-	4.96E+02 (1.95E+01)-
F_{b3}	2.14E-13 (7.53E-15)	2.36E-14 (4.09E-15)-	2.87E-13 (5.63E-14)+	3.57E-09 (4.03E-09)+	6.24E-14 (3.25E-15)-	1.95E+00 (3.78E-01)+	1.42E+01 (4.12E-01)+	9.38E-14 (1.56E-14)-	1.24E-13 (3.27E-15)-
F_{b4}	5.76E+10 (1.45E+10)	5.02E+11 (1.05E+11)+	1.23E+11 (8.63E+10)+	7.53E+11 (1.56E+11)+	3.68E+11 (2.53E+10)+	2.96E+11 (3.28E+10)+	4.53E+11 (8.56E+10)+	6.23E+11 (6.02E+10)+	3.67E+11 (5.81E+10)+
F_{b5}	3.30E+06 (2.59E+06)	2.67E+07 (7.23E+06)+	1.16E+07 (3.72E+06)+	2.75E+06 (1.89E+06)=	9.56E+06 (3.25E+06)+	3.74E+07 (4.36E+06)+	9.36E+07 (9.13E+06)+	1.37E+07 (2.03E+06)+	1.12E+07 (2.34E+06)+
F_{b6}	1.23E-07 (3.94E-09)	1.32E-01 (1.24E-01)+	2.02E+01 (1.65E-01)+	8.34E-07 (2.96E-08)+	1.95E-07 (8.25E-09)+	1.83E+01 (3.24E+00)+	3.83E+01 (1.52E+01)+	4.82E-02 (2.07E-02)+	6.34E-10 (4.18E-11)-
F_{b7}	6.59E+03 (1.16E+04)	8.67E+02 (2.12E+03)-	8.76E+04 (6.83E+04)+	2.13E+04 (2.53E+03)+	3.05E-04 (1.85E-04)-	6.82E+04 (5.53E+04)+	3.51E+04 (1.25E+05)+	3.63E-01 (2.82E-01)-	1.47E-01 (1.36E-01)-
F_{b8}	2.94E+04 (9.39E+03)	4.27E+07 (5.43E+06)+	2.86E+04 (9.46E+03)=	3.93E+07 (8.23E+05)+	3.13E+06 (2.82E+05)+	7.86E+06 (1.85E+06)+	2.13E+07 (1.93E+06)+	1.18E+07 (1.26E+07)+	1.06E+06 (1.25E+06)+
F_{b9}	6.13E+06 (5.41E+02)	5.28E+07 (3.18E+06)+	2.57E+07 (2.05E+06)+	7.35E+07 (5.93E+06)+	2.13E+07 (1.25E+06)+	3.51E+07 (4.53E+06)+	2.24E+07 (1.63E+06)+	3.34E+07 (1.52E+06)+	2.58E+07 (1.75E+06)+
F_{b10}	7.14E+03 (5.24E+01)	8.85E+02 (6.16E+01)+	2.56E+03 (1.88E+02)-	9.81E+03 (7.53E+01)+	5.63E+02 2.87E+01-	2.89E+03 (2.38E+02)-	5.28E+03 (4.36E+02)-	8.72E+02 (6.05E+01)-	7.28E+02 (4.76E+01)-
F_{b11}	1.01E+01 (9.36E-01)	1.05E+01 (1.47E+00)=	2.13E+01 (3.21E+00)+	4.65E-08 (6.03E-09)-	1.32E-12 (3.65E-13)-	2.86E+01 (2.46E+00)+	1.73E+02 (2.13E+00)+	2.11E+00 (2.52E+00)-	9.23E-14 (3.25E-13)-
F_{b12}	1.32E+02 (1.26E+02)	1.56E+04 (1.16E+04)+	1.46E+03 (7.36E+02)+	4.75E+05 (6.33E+04)+	4.61E+03 (3.92E+02)+	1.83E+04 (9.23E+03)+	2.42E+01 (9.93E+00)-	1.98E+04 (1.58E+03)+	1.97E+04 (2.15E+03)+
F_{b13}	9.91E+01 (9.95E+00)	7.48E+02 (1.23E+02)+	5.63E+02 (1.52E+02)+	6.65E+02 (2.13E+02)+	6.35E+02 (1.78E+02)+	9.63E+02 (3.15E+02)+	1.16E+05 (6.31E+04)+	9.82E+02 (2.05E+02)+	5.13E+02 (1.92E+02)+
F_{b14}	1.08E+07 (7.76E+05)	2.67E+08 (7.12E+06)+	7.03E+07 (4.86E+06)+	2.53E+08 (1.96E+07)+	6.68E+07 (2.68E+06)+	8.56E+07 (6.53E+06)+	1.23E+07 (1.86E+06)+	2.18E+09 (3.25E+07)+	6.58E+07 (4.05E+06)+
F_{b15}	8.98E+02 (4.62E+01)	8.25E+02 (3.61E+01)+	2.63E+03 (1.84E+02)+	1.23E+04 (5.68E+0)+	9.68E+03 (6.38E+01)+	1.52E+04 (8.36E+01)+	5.56E+03 (3.84E+02)+	8.56E+02 (5.26E+01)-	8.87E+02 (4.52E+01)=
F_{b16}	3.71E+00 (2.35E+00)	5.16E+00 (3.08E+00)+	7.35E+00 (7.32E+00)=	5.83E-8 (3.69E-9)-	1.86E-12 (2.68E-13)-	2.93E+01 (2.63E+01)+	3.63E+02 (7.25E+00)+	1.65E+00 (2.84E+00)-	5.18E-13 (6.87E-14)-
F_{b17}	8.07E+02 (7.88E+02)	9.32E+04 (3.13E+03)+	2.63E+04 (2.86E+03)+	2.63E+06 (1.63E+05)+	5.96E+04 (2.16E+03)+	9.03E+04 (1.63E+04)+	4.63E+01 (1.25E+01)-	5.39E+04 (2.58E+03)+	2.38E+05 (3.54E+04)+
F_{b18}	6.92E+02 (1.67E+02)	2.48E+03 (9.03E+02)+	1.59E+03 (2.05E+02)+	1.79E+03 (6.30E+02)+	1.78E+03 (2.67E+02)+	2.75E+03 (7.03E+02)+	2.53E+04 (1.35E+04)+	2.18E+03 (5.06E+02)+	1.18E+03 (3.07E+02)+
F_{b19}	1.23E+05 (1.12E+04)	1.83E+06 (9.82E+04)+	8.26E+05 (4.86E+04)+	1.36E+07 (5.38E+05)+	1.36E+06 (4.02E+04)+	6.03E+06 (7.38E+05)+	3.16E+06 (1.57E+05)+	8.93E+05 (1.18E+03)+	6.38E+06 (3.29E+05)+
F_{b20}	8.68E+02 (2.19E+00)	1.87E+03 (1.82E+02)+	1.84E+03 (2.03E+02)+	1.26E+03 (1.56E+02)+	1.33E+03 (8.54E+01)+	1.89E+03 (2.68E+02)+	9.92E+02 (1.58E+01)+	2.13E+03 (3.28E+02)+	9.06E+02 (2.58E+01)+
+/-/=		15/3/2	16/2/2	17/2/1	13/6/1/	19/1/0	17/3/0	13/7/0	11/8/1

4.4. Comparison with state-of-the-art algorithms for other non-PSO variants

In this subsection, HLPPO is compared with six other non-PSO variants to further assess its convergence performance. Three algorithms are chosen for comparison from the cooperative coevolution framework (MLCC [28], DECC-G [26], and DECC-DG [15]), one DE variant (DCDE [23]), one CS variant (CSRSA [23]), and the evolutionary competition champion algorithm (MASWCHAINS [47]). As shown in Table 4, HLPPO outperforms MLCC and DECC-G on at least 14 benchmarks, with only one fully separable function (F_{c2}) losing to MLCC. When compared to CSRSA, DECC-DG, and DCDE, HLPPO outperforms across the board in 15 benchmarks. In comparison to MASWCHAINS, HLPPO performs worse in only one separable benchmark (F_{c2}) and equally in one non-separable

Table 3

Experimental statistics of HLPPO and other PSO variants in the 1000-dimensional CEC 2013 benchmark test suite. The best average results for each benchmark are highlighted in bold.

Fun	HLPPO	DLLSO	AGLDPSO	CSO	TPLSO	SL-PSO	DMS-L-PSO	RLLPSO	SinSLSO
	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)
F_{c1}	7.53E-19 (2.27E-20)	3.88E-22 (1.43E-22) ⁻	2.14E-21 (1.43E-22) ⁻	2.86E-11 (2.28E-12) ⁺	6.23E-19 (3.05E-20) ⁼	1.35E-17 (3.06E-18) ⁺	1.85E+09 (1.31E+08) ⁺	8.78E-22 (3.68E-22) ⁻	1.47E-21 (1.38E-22) ⁻
F_{c2}	1.64E+3 (1.03E+2)	1.33E+03 (6.13E+01) ⁻	2.89E+03 (1.85E+02) ⁺	8.87E+3 (1.69E+2) ⁺	8.93E+02 (3.26E+01) ⁻	2.41E+03 (2.16E+02) ⁺	8.85E+03 (5.23E+02) ⁺	1.14E+03 (7.85E+01) ⁻	7.85E+02 (2.84E+01) ⁻
F_{c3}	1.76E-13 (1.26E-14)	2.18E+01 (1.36E-02) ⁺	2.11E+01 (2.25E-02) ⁺	2.32E+1 (1.42E-2) ⁺	2.36E+01 (4.36E-02) ⁺	2.34E+01 (3.65E-01) ⁺	2.15E+01 (2.05E-01) ⁺	2.23E+01 (6.34E-3) ⁺	2.19E+01 (6.05E-3) ⁺
F_{c4}	3.64E+8 (9.39E+7)	6.75E+09 (2.06E+09) ⁺	2.13E+09 (1.15E+08) ⁺	1.30E+10 (9.66E+8) ⁺	6.14E+09 (7.38E+08) ⁺	4.53E+09 (8.63E+08) ⁺	3.02E+11 (7.05E+10) ⁺	4.63E+09 (3.86E+08) ⁺	3.06E+09 (3.95E+08) ⁺
F_{c5}	5.23E+5 (1.56E+5)	6.63E+05 (1.58E+05) ⁼	3.23E+05 (4.51E+04) ⁻	4.21E+5 (9.45E+4) ⁻	5.63E+05 (1.82E+05) ⁼	6.37E+05 (1.36E+05) ⁼	4.02E+06 (5.77E+05) ⁺	7.23E+05 (1.13E+05) ⁺	5.36E+05 (9.36E+04) ⁺
F_{c6}	9.57E+0 (8.86E+0)	2.34E+06 (1.30E+03) ⁺	1.54E+06 (1.95E+03) ⁺	2.86E+6 (1.36E+5) ⁺	1.23E+06 (1.32E+03) ⁺	1.37E+06 (2.05E+03) ⁺	1.23E+06 (3.42E+04) ⁺	1.23E+06 (8.26E+02) ⁺	1.29E+06 (1.14E+03) ⁺
F_{c7}	6.59E+5 (5.21E+4)	1.53E+06 (8.35E+05) ⁺	4.15E+06 (1.23E+06) ⁺	1.26E+7 (1.91E+6) ⁺	2.35E+05 (2.31E+04) ⁻	1.97E+06 (8.06E+05) ⁺	1.45E+09 (6.87E+08) ⁺	1.56E+05 (2.04E+05) ⁻	9.86E+05 (2.38E+05) ⁺
F_{c8}	2.57E+13 (1.78E+12)	1.34E+14 (3.87E+13) ⁺	1.32E+13 (4.92E+12) ⁻	2.03E+14 (2.39E+13) ⁺	3.15E+14 (3.06E+13) ⁺	1.85E+14 (4.12E+13) ⁺	2.89E+14 (2.56E+14) ⁺	8.18E+13 (1.24E+13) ⁺	5.86E+13 (1.70E+13) ⁺
F_{c9}	3.92E+7 (3.09E+6)	1.38E+08 (3.75E+07) ⁺	1.95E+08 (4.58E+07) ⁺	3.04E+7 (1.68E+6) ⁼	3.91E+07 (3.56E+06) ⁼	8.63E+07 (3.05E+07) ⁺	3.85E+08 (4.58E+07) ⁺	3.83E+07 (7.28E+06) ⁻	3.15E+07 (6.13E+06) ⁻
F_{c10}	2.26E+3 (3.03E+3)	9.27E+07 (3.61E+05) ⁺	9.45E+07 (4.23E+05) ⁺	8.61E+7 (5.86E+6) ⁺	9.15E+07 (2.36E+05) ⁺	9.17E+07 (1.52E+06) ⁺	9.65E+07 (2.01E+06) ⁺	9.45E+07 (2.04E+05) ⁺	9.43E+07 (1.05E+05) ⁺
F_{c11}	4.68E+6 (2.84E+6)	9.21E+11 (6.85E+09) ⁺	9.38E+11 (4.62E+10) ⁺	1.82E+11 (1.14E+9) ⁺	2.38E+08 (3.86E+07) ⁺	9.26E+11 (2.08E+10) ⁺	1.25E+11 (6.58E+10) ⁺	9.26E+11 (8.26E+09) ⁺	1.05E+08 (8.93E+06) ⁺
F_{c12}	9.50E+2 (2.62E-10)	1.89E+03 (1.35E+02) ⁺	1.55E+03 (2.86E+02) ⁺	1.04E+3 (5.61E-1) ⁺	1.53E+03 (9.36E+01) ⁺	1.84E+03 (1.98E+02) ⁺	6.58E+04 (5.38E+04) ⁺	1.82E+03 (1.84E+02) ⁺	1.14E+03 (3.25E+01) ⁺
F_{c13}	5.38E+6 (3.88E+6)	3.56E+08 (1.16E+08) ⁺	3.02E+10 (5.23E+09) ⁺	6.46E+8 (1.58E+8) ⁺	9.84E+07 (4.15E+07) ⁺	4.88E+08 (1.84E+08) ⁺	1.85E+10 (6.35E+09) ⁺	2.09E+08 (3.57E+07) ⁺	1.27E+08 (6.28E+07) ⁺
F_{c14}	1.76E+7 (2.63E+6)	1.92E+08 (1.13E+08) ⁺	9.56E+07 (4.46E+06) ⁺	5.35E+9 (2.70E+9) ⁺	2.98E+07 (4.62E+06) ⁺	3.57E+08 (2.85E+08) ⁺	2.48E+11 (1.57E+11) ⁺	5.21E+07 (1.89E+07) ⁺	1.38E+08 (4.36E+07) ⁺
F_{c15}	5.89E+6 (1.31E+5)	5.24E+06 (3.85E+05) ⁼	2.89E+06 (3.57E+05) ⁻	7.39E+7 (6.57E+6) ⁺	2.38E+06 (1.31E+05) ⁻	5.93E+07 (6.47E+06) ⁺	2.04E+07 (3.85E+06) ⁺	1.56E+06 (6.05E+04) ⁻	1.68E+07 (1.06E+06) ⁺
+/-/=		12/2/1	11/4/0	13/1/1	9/3/3	14/0/1	15/0/0	11/4/0	12/3/0

benchmark (F_{c15}), while succeeding in 13 benchmarks. HLPPO's excellent optimization results can be attributed to a variety of factors, including particle classification into positive, negative, or homogenized categories, each with unique characteristics and treatments. Furthermore, the proposed homogenization learning concept allows each particle to learn from three distinct subpopulations, enhancing its exploitation potential by learning from the superior positive particle subpopulation and exploring potential by learning from the inferior negative particle subpopulation. In high-dimensional optimization, the homogeneous learning term also regulates the learning ability of subpopulations at different evolutionary stages, resulting in a more desirable balance between exploration and exploitation. The homogeneous learning strategy increases the diversity of the particle population by identifying the dynamic diversity of subpopulations to guide the evolutionary direction. This increases the possibility of jumping out of the local optimum. During the positive particle learning phase, particles learn from each other with a narrower learning range towards the dominant positive particles, increasing the potential for exploitation. As a result, as shown in Tables 2, 3, and 4, HLPPO can better balance algorithm exploration and exploitation and compete with state-of-the-art algorithms.

The CEC 2013 benchmarks are used to compare the computation times of HLPPO and fourteen other competitive algorithms, including eight PSO variants and six non-PSO variants, to evaluate the efficiency and applicability of the algorithms in practical applications. The average computation times for the fifteen algorithms used to solve the 1000-dimensional CEC 2013 benchmark are shown in Table 5, which is run 30 times independently. Each run has a fitness evaluation count of $3000 \times D$.

TPLSO employs both group and elite learning, with a competitive mechanism used to update learning group members in the population learning phase. During the elite learning phase, particles in the swarm are sorted based on their fitness values. Because both phases necessitate sorting, TPLSO's computation time will inevitably increase. Furthermore, DLLSO's hierarchical learning mechanism is similar, and as shown in Table 5, it has a longer running time. DECC-DG, DECC-G, and MLCC, on the other hand,

Table 4

Experimental statistics of HLPPO and other non-PSO variants in the 1000-dimensional CEC 2013 benchmark test suite. The best average results for each benchmark are highlighted in bold.

Fun	HLPPO	CSRSA	DCDE	MASWCHAINS	DECC-DG	DECC-G	MLCC
	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)	Mean (Std)
F_{c1}	7.53E-19 (2.27E-20)	2.34E+11 (2.34E+11)+	9.55E+06 (1.80E+03)+	8.49E-13 (2.18E-13)+	1.38E+06 (1.63E+06)+	2.85E-6 (3.05E-06)+	9.35E-14 (2.11E-11)+
F_{c2}	1.64E+03 (1.03E+02)	6.06E+04 (1.86E+03)+	7.63E+03 (4.46E+02)+	1.22E+03 (2.28E+01)-	2.05E+05 (1.37E+04)+	1.51E+03 (2.36E+02) =	3.67E+00 (1.24E+00)-
F_{c3}	1.76E-13 (1.26E-14)	2.13E+01 (1.33E-02)+	1.43E+01 (5.36E-01)+	2.14E+01 (1.12E-02)+	2.38E+01 (3.28E-01)+	2.05E+01 (4.31E-02)+	2.85E+00 (1.53E-01)+
F_{c4}	3.64E+08 (9.39E+07)	3.17E+13 (1.10E+13)+	2.25E+10 (1.01E+10)+	4.58E+09 (4.91E+08)+	5.39E+11 (2.25E+11)+	2.81E+11 (1.56E+11)+	2.58E+11 (1.27E+11)+
F_{c5}	5.23E+05 (1.56E+05)	4.86E+07 (2.57E+06)+	1.58E+06 (1.38E+06)+	1.87E+06 (6.13E+04)+	5.94E+06 (4.86E+05)+	8.96E+06 (1.56E+06)+	1.63E+07 (3.51E+06)+
F_{c6}	9.57E+00 (8.86E+00)	1.04E+06 (3.70E+03)+	1.91E+01 (5.82E-01)+	1.01E+06 (3.06E+03)+	2.13E+06 (1.87E+03)+	2.34E+06 (2.57E+03)+	1.25E+06 (5.26E+03)+
F_{c7}	6.59E+05 (5.21E+04)	7.25E+14 (3.46E+14)+	7.58E+07 (6.32E+07)+	3.45E+06 (2.53E+05)+	1.38E+09 (6.85E+08)+	1.25E+09 (2.15E+08)+	1.56E+09 (1.05E+09)+
F_{c8}	2.57E+13 (1.78E+12)	1.44E+18 (4.46E+17)+	1.69E+14 (7.92E+13)+	4.85E+13 (2.03E+12)+	8.56E+15 (3.67E+15)+	7.89E+15 (3.06E+15)+	9.87E+15 (5.98E+15)+
F_{c9}	3.92E+07 (3.09E+06)	4.19E+09 (5.31E+08)+	1.09E+08 (2.70E+07)+	1.07E+08 (3.36E+06)+	1.05E+09 (1.03E+08)+	6.17E+08 (8.35E+07)+	9.63E+08 (3.05E+08)+
F_{c10}	2.26E+03 (3.03E+03)	9.43E+07 (8.62E+05)+	2.97E+02 (6.86E+01)+	9.18E+07 (2.12E+05)+	9.03E+07 (2.84E+05)+	8.95E+07 (3.16E+05)+	9.37E+07 (1.02E+06)+
F_{c11}	4.68E+06 (2.84E+06)	2.82E+16 (1.21E+16)+	3.86E+10 (2.48E+10)+	2.19E+08 (5.96E+06)+	7.93E+11 (6.13E+10)+	1.53E+11 (5.14E+10)+	3.15E+11 (1.26E+11)+
F_{c12}	9.50E+02 (2.62E-10)	2.96E+12 (7.80E+10)+	3.12E+07 (1.22E+08)+	1.25E+03 (2.11E+01)+	6.53E+10 (5.38E+09)+	4.86E+03 (8.35E+02)+	2.63E+03 (6.85E+02)+
F_{c13}	5.38E+06 (3.88E+06)	4.63E+16 (1.91E+16)+	4.29E+09 (1.19E+09)+	1.98E+07 (3.64E+05)+	5.84E+10 (2.76E+09)+	9.87E+09 (2.53E+09)+	1.34E+10 (3.96E+09)+
F_{c14}	1.76E+07 (2.63E+06)	7.74E+16 (4.23E+16)+	6.37E+10 (2.23E+10)+	1.36E+08 (4.22E+06)+	7.74E+11 (6.32E+10)+	1.75E+11 (6.05E+10)+	2.41E+11 (6.24E+10)+
F_{c15}	5.89E+06 (1.31E+05)	1.04E+16 (2.86E+15)+	9.93E+06 (3.02E+06)+	5.71E+06 (1.51E+05) =	8.59E+07 (5.72E+06)+	1.32E+07 (2.14E+06)+	1.94E+07 (2.23E+06)+
+/-/=		15/0/0	15/0/0	13/1/1	15/0/0	14/0/1	14/1/0

are non-PSO variants that rely on the cooperative evolution principle. The optimization process requires a significant amount of resources to handle the high-dimensional decomposition of multiple low-dimensional problems, resulting in significant increases in computation time. TPLSO, DLLSO, and other optimization algorithms with cooperative evolution mechanisms have longer average computation times than HLPPO. This is because HLPPO only needs to deal with fitness ranking once, and particle fitness evaluations are only required for one-third of the population size. Although CSO, SL-PSO, SinSLSO, and DCDE have shorter running times than HLPPO, HLPPO is preferred due to its superior performance.

Furthermore, the Friedman test and post-hoc Nemenyi non-parametric statistical tests [48] are performed at the significance level of 0.05 to confirm the difference between HLPPO and other algorithms, and the corresponding p -values for each algorithm are shown in Table 6. The results show significant differences between HLPPO and the comparison algorithms.

4.5. Comparisons on ultra-high-dimensional functions

To investigate HLPPO's search ability on ultra-high-dimensional functions, additional experiments are performed on F_{a1} to F_{a6} in 2000 dimensions of the CEC 2008 test suite. Because its global optimum is dimensionally dependent and its scalability is difficult to assess, F_{a7} is excluded. F_{a5} is classified as a separable function in ultra-high dimensions, where the effect of the product part becomes less important. Separable functions are F_{a1} , F_{a4} , F_{a5} , and F_{a6} , whereas non-separable functions are F_{a2} and F_{a3} . TPLSO and CSO are two PSO variants chosen for comparison with HLPPO. A non-parametric test of a two-tailed Wilcoxon rank sum is used to compare these optimization algorithms at the 0.05 level of significance.

Table 5
Average computation time of HLPSON and competitive algorithms.

Algorithm	Average computation time (in seconds)
DLLSO	101.26
AGLDPSO	92.17
CSO	58.23
TPLSO	121.07
SL-PSO	75.39
DMS-L-PSO	93.28
RLLPSO	84.27
SinSLSO	78.28
CSRSA	92.05
DCDE	65.83
MASWCHAINS	86.35
DECC-DG	116.61
DECC-G	113.45
MLCC	94.32
HLPSON	79.88

Table 6
Friedman test and post-hoc Nemenyi statistical validation for the CEC 2013 benchmark test suite with HLPSON as the control algorithm.

Algorithm	<i>p</i> -value
DLLSO	1.83E-04
AGLDPSO	2.34E-05
CSO	1.62E-04
TPLSO	2.14E-04
SL-PSO	5.13E-05
DMS-L-PSO	2.13E-06
RLLPSO	5.02E-04
SinSLSO	3.31E-05
CSRSA	2.39E-06
DCDE	1.72E-08
MASWCHAINS	4.21E-07
DECC-DG	3.09E-09
DECC-G	2.18E-08
MLCC	1.28E-10

Tables 7 and 8 demonstrate the results of the six CEC 2008 benchmarks for HLPSON, TPLSO, and CSO, which represent the average of 20 independent runs. Table 7 shows that the algorithm consumes $3.00\text{E}+05$ fitness evaluations per run, while Table 8 consumes $3.00\text{E}+06$ fitness evaluations with $N_p = 1008$. Overall, HLPSON, TPLSO, and CSO are optimized better in F_{a1} , F_{a4} , F_{a5} , and F_{a6} than in F_{a2} and F_{a3} . This is primarily due to the fact that non-separable functions necessitate more diverse information in the optimization process, which is especially important in ultra-high-dimensional problems. TPLSO achieves better initial results for F_{a4} , F_{a5} , and F_{a6} , but converges slowly and tends to fall into a local optimum when more fitness evaluations are used, as shown in Table 6. CSO, on the other hand, is superior for optimization in these three benchmarks due to its potentially greater diversity.

In addition, as observed in Tables 7 and 8, HLPSON outperforms TPLSO and CSO in all six benchmarks and converges faster. Unlike other special optimization strategies that specialize in solving large-scale problems, traditional PSO is extremely difficult to optimize for ultra-high-dimensional problems with more than 1000 dimensions. Fortunately, HLPSON makes use of the concept of homogeneous learning, as well as dynamic subpopulation identification methods and autophagy mechanisms, allowing it to demonstrate its superior search ability when dealing with complex, ultra-high-dimensional problems.

4.6. Comparisons on low-dimensional functions

To investigate the convergence of HLPSON on low-dimensional functions, experiments are conducted on the 30-dimensional functions F_{d1} to F_{d30} in the CEC 2017 benchmark suite. Because of its high randomness in function values, F_{d2} is excluded. As presented in Table 9, good convergence performance has been achieved by HLPSON compared with five other excellent PSO variants (TLS-PSO [10], ADFPSO [12], SL-PSO, PSO-CL [11], and HPSO-FDB-ISE [9]) and three non-PSO variants (CMAES [30], LSHADE [49], and COA [50]). HLPSON demonstrates exceptional performance on F_{d6} and F_{d9} , particularly in finding the theoretical global optimum on F_{d9} . TLS-PSO finds the global optimum on F_{d9} as well, but its search performance on hybrid and composite functions is inferior to HLPSON. LSHADE handles separable problems well, and the theoretical global optimum has been found on F_{d1} , F_{d2} , and F_{d9} , but its search performance on composite functions is inferior to HLPSON. Notably, F_{d30} is a highly challenging function with multimodal, non-

Table 7

Experimental statistics of HLP SO, TPL SO, and CSO in the 2000-dimensional CEC 2008 benchmark test suite for the fitness evaluations of $3.00\text{E}+05$ are presented. The best average results for each benchmark have been marked in bold.

Fun	HLP SO	TP L SO	CSO	Global optimum
	Mean (Std)	Mean (Std)	Mean (Std)	
F_{a1}	1.03E+03 (2.86E+01)	4.59E+06 (6.56E+04)+	6.02E+05 (9.08E+03)+	0
F_{a2}	8.93E+01 (1.45E+00)	1.10E+02 (2.86E+00)+	1.25E+02 (1.13E+00)+	0
F_{a3}	2.48E+07 (2.37E+06)	1.59E+12 (3.97E+10)+	1.41E+11 (9.10E+09)+	0
F_{a4}	1.31E+04 (4.59E+02)	2.99E+04 (3.68E+02)+	2.33E+05 (1.54E+03)+	0
F_{a5}	1.01E+01 (4.80E-01)	4.07E+02 (2.78E+02)+	5.17E+03 (7.98E+01)+	0
F_{a6}	1.97E+00 (2.54E-02)	2.05E+01 (2.48E-02)+	1.50E+01 (7.31E-02)+	0
+/-/=		6/0/0	6/0/0	

Table 8

Experimental statistics of HLP SO, TPL SO, and CSO in the 2000-dimensional CEC 2008 benchmark test suite for the fitness evaluations of $3.00\text{E}+06$ are presented. The best average results for each benchmark have been marked in bold.

Fun	HLP SO	TP L SO	CSO	Global optimum
	Mean (Std)	Mean (Std)	Mean (Std)	
F_{a1}	1.54E-22 (3.27E-23)	4.47E+06 (6.05E+04)+	1.67E+02 (7.38E+01)	0
F_{a2}	6.07E+01 (1.23E+00)	1.07E+02 (2.16E+00)+	9.28E+01 (5.28E+00)+	0
F_{a3}	1.93E+03 (3.28E+02)	1.58E+12 (3.88E+10)+	3.67E+8 (2.14E+07)+	0
F_{a4}	2.42E+03 (4.59E+02)	2.99E+04 (2.18E+02)+	2.41E+04 (3.87E+02)+	0
F_{a5}	3.33E-16 (1.15E-16)	4.05E+02 (3.21E+01)+	2.41E+00 (1.56E+00)+	0
F_{a6}	3.35E-13 (6.17E-12)	2.05E+01 (2.05E-02)+	1.07E+00 (1.12E-02)+	0
+/-/=		6/0/0	6/0/0	

separable, and asymmetric properties as well as different attribute combinations of subcomponents around different local optima. HLP SO, however, is able to tackle such a difficult function by outperforming the other eight excellent algorithms in terms of search performance. To verify the difference between HLP SO and other algorithms, the Friedman test and post-hoc Nemenyi non-parametric statistical test are performed at a significance level of 0.05, as shown in Table 10. The results indicate significant differences between HLP SO and the comparison algorithms, demonstrating that HLP SO not only performs well on ultra-high-dimensional optimization problems, but also on low-dimensional optimization problems.

5. Conclusion

In this paper, we propose HLP SO, a novel high-dimensional global optimization algorithm that effectively addresses the dimensionality curse. The algorithm incorporates the concept of homogeneous learning, which results in a better balance of exploration and exploitation. To improve the convergence speed of high-dimensional optimization, a dynamic subpopulation diversity measure and a method for identifying population diversity are proposed. An evolutionary strategy involving an autophagy mechanism is also presented. A comparison of the CEC 2008, CEC 2010, CEC 2013, and CEC 2017 test suites demonstrates that the proposed HLP SO outperforms several state-of-the-art high-dimensional and low-dimensional optimization algorithms despite its simplicity.

Table 9

Experimental statistics of HLPPO and other competitive algorithms in the 30-dimensional CEC 2017 benchmark test suite. The best average results for each benchmark are highlighted in bold.

Fun	HLPPO	TLS-PSO	ADFPPO	SL-PSO	PSO-CL	HPSO-FDB-ISE	CMAES	LSHADE	COA
	Mean	Mean	Mean	Mean	Mean	Mean	Mean	Mean	Mean
F_{d1}	2.52E+03	2.24E+03	1.86E+03	4.76E+03	1.81E+03	2.86E+03	5.41E+4	0.00E+00	4.51E+03
F_{d3}	5.34E+01	9.20E+00	3.60E+14	8.85E+03	3.17E-08	3.56E+02	3.24E+5	0.00E+00	4.12E+02
F_{d4}	4.68E+01	1.12E+02	4.82E+01	9.13E+01	2.53E+01	4.93E+02	4.51E+2	5.88E+01	5.13E+02
F_{d5}	5.93E+00	1.13E+02	/	2.23E+01	4.38E+01	5.40E+02	5.49E+2	6.77E+00	5.89E+02
F_{d6}	9.85E-14	1.51E-07	/	2.28E-06	4.88E-04	6.00E+02	6.16E+2	1.55E-08	6.98E+02
F_{d7}	3.51E+01	4.70E+01	/	1.98E+02	7.58E+01	7.68E+02	9.77E+2	3.78E+01	8.02E+02
F_{d8}	6.36E+00	1.03E+01	/	2.12E+01	4.58E+01	8.40E+02	8.63E+2	7.98E+00	9.13E+02
F_{d9}	0.00E+00	0.00E+00	/	1.14E-01	2.78E+00	9.01E+02	9.12E+2	0.00E+00	1.24E+03
F_{d10}	8.93E+02	1.39E+03	/	9.11E+02	2.74E+03	4.21E+03	1.64E+4	1.48E+03	1.44E+03
F_{d11}	9.32E+00	5.93E+01	3.55E+01	2.45E+01	3.85E+01	1.16E+03	6.73E+4	3.52E+01	1.02E+03
F_{d12}	1.76E+04	1.60E+04	2.71E+05	4.98E+04	9.63E+04	1.08E+05	1.23E+7	1.00E+03	1.35E+04
F_{d13}	8.58E+03	7.83E+03	2.90E+04	1.65E+04	2.39E+03	2.03E+04	9.55E+6	1.63E+01	1.35E+03
F_{d14}	6.99E+03	1.14E+03	1.58E+03	2.26E+04	1.10E+04	6.92E+03	8.46E+5	2.27E+01	1.43E+03
F_{d15}	1.34E+03	2.05E+03	2.22E+04	4.56E+03	1.49E+03	8.79E+03	4.65E+6	3.76E+00	1.52E+03
F_{d16}	1.63E+02	3.01E+02	3.61E+02	2.65E+02	4.52E+02	2.02E+03	9.19E+3	5.84E+01	1.67E+03
F_{d17}	5.24E+01	8.30E+01	1.08E+02	6.34E+01	1.12E+02	1.85E+03	9.78E+3	5.89E+01	1.68E+03
F_{d18}	6.35E+04	6.15E+04	7.74E+04	1.28E+05	9.78E+04	1.97E+05	5.47E+6	5.83E+01	1.92E+03
F_{d19}	2.96E+03	3.57E+03	5.77E+03	4.88E+03	3.33E+03	5.33E+03	5.32E+5	1.02E+01	1.92E+03
F_{d20}	1.26E+02	1.95E+02	1.84E+02	1.19E+02	1.27E+02	2.23E+03	9.92E+3	3.87E+01	2.15E+03
F_{d21}	2.01E+02	2.15E+02	2.22E+02	2.21E+02	2.48E+02	2.34E+03	9.13E+3	2.02E+02	2.18E+03
F_{d22}	1.59E+02	1.00E+02	1.00E+02	1.50E+02	1.07E+02	2.30E+03	1.80E+4	1.00E+02	2.31E+03
F_{d23}	3.61E+02	3.58E+02	3.63E+02	3.87E+02	4.04E+02	2.69E+03	9.27E+3	3.61E+02	2.71E+03
F_{d24}	4.31E+02	4.23E+02	4.31E+02	4.51E+02	4.78E+02	2.87E+03	9.88E+3	4.37E+02	2.74E+03
F_{d25}	3.87E+02	3.88E+02	3.87E+02	3.87E+02	3.87E+02	2.89E+03	9.39E+3	3.87E+02	2.35E+13
F_{d26}	1.14E+03	4.12E+02	4.15E+02	1.22E+03	1.30E+03	3.57E+03	1.56E+4	9.96E+02	2.98E+03
F_{d27}	5.00E+02	5.26E+02	5.07E+02	5.20E+02	5.08E+02	3.21E+03	9.54E+3	5.09E+02	3.12E+03
F_{d28}	3.72E+02	3.69E+02	4.17E+02	3.64E+02	3.39E+02	3.20E+03	9.83E+3	3.45E+02	3.32E+03
F_{d29}	4.38E+02	4.65E+02	4.69E+02	5.36E+02	5.58E+02	3.49E+03	1.41E+4	4.39E+02	3.13E+03
F_{d30}	1.87E+03	5.73E+03	7.73E+03	4.78E+03	3.76E+03	8.55E+03	9.92E+5	2.00E+03	3.12E+04
+/-/=		7/21/1	4/18/1	3/25/1	6/22/1	1/28/0	0/29/0	13/13/3	4/25/0

Table 10

Friedman test and post-hoc Nemenyi statistical validation for the CEC 2017 benchmark test suite with HLPPO as the control algorithm.

Algorithm	<i>p</i> -value
TLS-PSO	3.06E-06
ADFPPO	1.21E-05
PSO-CL	4.08E-05
HPSO-FDB-ISE	5.32E-06
CMAES	3.07E-11
LSHADE	2.83E-07
COA	2.26E-06
DECC-DG	3.09E-09
DECC-G	2.18E-08
MLCC	1.28E-10

With the increasing use of artificial intelligence in information science, the emergence of large scale optimization problems warrants further investigation in the future. To begin, more advanced mechanisms can be developed to better update the control parameters of HLPPO, thereby improving the algorithm's robustness. Second, it would be interesting to investigate the application of the PSO homogenization learning concept to multi-objective optimization problems, such as the design of analog integrated circuits and routing problems. Furthermore, it is critical to investigate the combination of PSO's autophagy mechanism and diversity identification techniques with other metaheuristic algorithms in order to fully realize its potential in neural network training and evolution. This will be a critical area of future study.

CRediT authorship contribution statement

Wen-Yuan Fu: Conceptualization, Methodology, Software, Writing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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